VICTORIA JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATIONS 2024

H2 GEOGRAPHY **9173/02**
Paper 2

3 hours

READ THESE INSTRUCTIONS FIRST

Write your name and index number on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer Question 1 in **Section A**.
Answer Question 2 in **Section B**.
Answer **one** question in **Section C**.

The Insert contains all the Resources referred to in the questions.
You should make reference to appropriate examples studied in the field or the classroom, even where such examples are not specifically requested by the question.
Diagrams and sketch map should be drawn whenever they serve to illustrate and answer.
You are reminded of the need for good English and clear presentation of your answers.

At the end of the examination fasten all your work securely together by section.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Cluster 4 Fieldwork

1 A group of four 18-year-old students in Singapore undertook a fieldwork investigation to understand the flood risk in Edogawa City in Tokyo, Japan. Edogawa City, a city with more than 695 000 people, is bounded by the Arakawa River in west and Edo River in the east and is in the downstream location of both rivers. The investigation took place during 4-7 June of their June holidays.

Prior to their trip to Japan, they researched for online secondary data provided by the Risk Management Department of Edogawa City for the flood hazard map in Edogawa City. They used this map to identify four sites to conduct their fieldwork in and crafted their hypothesis based on that.

They planned to investigate the river velocity, channel width, channel depth. They also planned to repeat the velocity readings 3 times then find the average. Students were provided the following equipment:

- Tape measure (50 metre)
- 10m twine x 2
- Metre ruler
- Stop-watch
- 5 oranges
- Ranging poles

However, during their fieldwork in Japan, they found that they were not able to conduct their river studies. Since they could not conduct it, they searched online for data on the river measurements that they required.

They continued with their second data collection method, which is to conduct surveys with 25 people at each of the four sites. They went to one site per day, from 2-4pm to conduct the survey. They used systematic sampling to identify the sample population and also ensured that there was a mix of people from different age groups and genders.

Resource 1 shows the four sites of investigation chosen by the students based on a flood hazard map by the city government. Resource 2 shows a ground view of one of the sites. Resource 3 show the river measurements of the four sites based on online sources. Resource 4 shows the questionnaire survey that the students used. Resource 5 shows the tabulated results of Question 3 from the questionnaire survey.

(a) With reference to Resource 1, suggest a suitable hypothesis and explain why it is suitable. [3]

(b) With reference to Resource 1, suggest reasons why the students chose to do fieldwork at these four sites. [4]

(c) Using Resource 2 and 3 only, suggest why it was not feasible to take river measurements at the chosen sites. [5]

(d) Describe the steps students could have taken to measure the river velocity with the given equipment. [6]

(e) Using Resource 4, explain the strengths and weaknesses in their questionnaire survey. [6]

(f) Using a suitable data representation method, present the results to Question 3 in the questionnaire survey shown in Resource 5, and justify your choice. [6]

(g) The students concluded that Site 1 has the highest flood risk. With reference to all resources, evaluate the validity of their conclusion. [10]

Section B

Cluster 3 Sustainable Future and Climate Change

2 The City Liveability Index developed by the Economist Intelligence Unit rates living conditions in 173 cities across five categories: stability, healthcare, culture and environment, education, and infrastructure. Vienna, Austria has been consistently ranking top or near the top for many years. The city is also known for its sustainability efforts and inclusive urban planning.

Resource 6 is a map of cities ranked by the City Liveability Index in 2024. Resource 7 shows the location of green spaces and metro (train) stations in Vienna. Resource 8 is an adapted article excerpt on climate protection areas in Vienna. Resource 9 shows inclusive urban designs in Vienna.

(a) With reference to Resource 6, describe the distribution of the world's cities based on their liveability. [4]

(b) Explain what makes a city liveable. [4]

(c) Describe the strengths and limitations of tools such as the City Liveability Index in measuring urban liveability. [6]

(d) Explain how anthropogenic factors have contributed to contemporary climate change. [6]

(e) With reference to Resource 7 and 8, describe how Vienna has successfully improved urban liveability while also mitigating climate change. [6]

(f) With reference to Resource 9, describe how Vienna's urban design meets the needs of different social groups within the city. [4]

Section C

Cluster 3 Sustainable Future and Climate Change

Answer either question 3 or question 4

3 Evaluate the success of slum management strategies in helping cities in countries at lower levels of development achieve sustainable urban development. [20]

4 'Sea level rise is the greatest threat of contemporary climate change.'

Evaluate this statement. [20]